PROFESSOR EMILE R. CHIMUSA

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PROFILE

Professor Chimusa is an internationally recognised leader in bioinformatics, computational biology and AI-driven health data science. He holds a UK Academy of Medical Sciences Professorship (2023), one of the most competitive individual research leadership awards in UK biomedicine, and has authored 112 peer-reviewed publications, a book and five book chapters spanning human genomics, disease risk prediction and health data science in underrepresented populations. A Senior Fellow of the Higher Education Academy (SFHEA), he combines this research leadership with sustained academic citizenship, programme leadership, and international capacity-building across Africa, Europe and North America, underpinned by long-standing expertise in reproducible, FAIR-aligned data science and open-source tool development.

CURRENT POSITIONS

2025 - present: Founder and Chair, AFROMIND (African Society for Omics Intelligence and Health Data Science) - a pan-African professional network advancing AI-driven omics and data science for equitable health outcomes.

2023 - present: The UK Academy of Medical Sciences Professor.

2023 - present: Visiting Professor, Population Health Sciences Institute, Newcastle University.

2022 - present: Professor of Bioinformatics and Computational Biology, Northumbria University.

2020 - present: Founder, Omics Data Solutions - genomic data science training and consulting company, registered in South Africa and Ireland, serving research and industry clients across 50+ countries and partnering with the African Academy of Sciences on bioinformatics capacity-building.

2016 - present: Coordinator, Bioinformatics Ecosystem, Centre for High-Performance Computing (CHPC), Cape Town, South Africa.

2015 - present: Visiting lecture, The African Institute for Mathematical Sciences (AIMS) and Next Einstein Forum (NEF), South Africa, Senegal, Ghana, Cameroon, Rwanda.

EDUCATION

2009 - 2013: PhD, Computational Biology, Division of Computational Biology, Department of Integrative Biomedical Sciences, University of Cape Town, South Africa; Visiting PhD student, Harvard T.H. Chan School of Public Health, USA. Thesis: "*Mapping genes underlying ethnic differences in tuberculosis risk by linkage disequilibrium in the South African Coloured population of the Western Cape.*" Supervisors: Profs Nicola Mulder, Alkes Price, Paul van Helden, Eileen Hoal.

2008 - 2009: MSc, Computational Biology (upgraded to PhD), University of Cape Town, South Africa.

2007 - 2008: Structured MSc / Postgraduate Diploma, Mathematical Sciences, African Institute for Mathematical Sciences (AIMS), University of Western Cape, South Africa.

1999 - 2005: BSc (Honours, First Class), Applied Mathematics and Informatics, University of Kinshasa, Democratic Republic of Congo.

ACADEMIC & PROFESSIONAL APPOINTMENTS

2021 - 2022: Programme Convener, Computational Health Informatics, Faculty of Health Sciences, University of Cape Town (as Associate Professor).

2019 - 2022: Associate Professor of Computational Biology, Division of Human Genetics, Department of Pathology, University of Cape Town.

2017 - 2021: Associate Head, Division of Human Genetics (Education & Learning), Department of Pathology, University of Cape Town; periodically Acting Head of Division.

2016 - 2022: Programme Convener, MSc/Honours in Human Genetics & Forensic Genetics, University of Cape Town.

2016 - 2019: Senior Lecturer, Division of Human Genetics, Department of Pathology, University of Cape Town.

2014 - 2016: Lecturer, Division of Computational Biology, University of Cape Town.

2013 - 2014: Postdoctoral Research Fellow, Division of Computational Biology, University of Cape Town; concurrently Biostatistical Scientific Researcher, Centre for Proteomic and Genomic Research (CPGR), Cape Town.

RESEARCH FUNDING (SELECTED)

Career research income exceeding £17.7 million as PI, Co-PI or Co-Investigator across national and international consortia (source: Northumbria Research Portal); selected grants below.

Northumbria University (2022 – present)

- PI - "Optimal Framework for Disease Risk Prediction and Stratification", Academy of Medical Sciences, £489,291.54 [awarded 2023].
- PI - "Orthoebolavirus Real-time Bio-surveillance Intelligence Twin (ORBIT)", Bill & Melinda Gates Foundation, £ 440,368.42 (2026, with Prof. Dav Ebeng, Institut National de Recherche Biomédicale (INRB), Democratic Republic of the Congo (DRC) [Awaiting official award announcement].
- Co-I - "Identifying and therapeutically targeting immune mediators of pathology in genetic mitochondrial disease", MRC, £1,418,830.10 (submitted 2026, with Prof. Simon Johnson, Dr Monika Winter; Dr. Simon Bridge and Dr Uwe Richter, Newcastle University).

Major International Grants (University of Cape Town-led / Consortium, Selected)

- PI - Computational Omics and Biomedical Informatics Program (COBIP), National Institutes of Health (NIH), $1,730,785 (2021 - 2022).
- Expert/Consultant - Developing a Sickle Africa Data Coordinating Center (SADaCC), NIH/NHGRI, $3,577,904 (2021 - 2025).
- Co-PI - H3ABioNet, Pan African Bioinformatics Network, NIH/NHGRI, $2,500,000 (2017 - 2023)
- Co-Investigator - Hearing Impairment Genetics Studies in Africa (HI-GENES Africa), Wellcome Trust/AESA, $2,067,742 (2017 - 2022).
- Co-PI - Hearing Impairment Genetics Studies in Africa (HI-GENES Africa), NIH/NHGRI, $1,249,097 (2017 – 2022).
- Co-Investigator - Individual Findings in Genetic Research in Africa (IFGeneRA), NIH/NHGRI, $2,499,081 (2017 - 2021).
- PI - Novel anticancer compounds targeting unusual nucleic acid structures in cancer therapy, South African Medical Research Council (BRICS multilateral), ZAR 1,500,000 (2021 - 2023)
- PI - CHPC National Flagship Project, Computing Ecosystems equipment award, CSIR South Africa, ZAR 2,850,000 (2018).
- Co-PI - Pharmacogenomics of tamoxifen in the treatment of breast cancer among South Africans, UCT ACC, ZAR 1,900,000 (2019 - 2022).
- PI - Computational method to understand population admixtures, International Development Research Centre (IDRC), Canada, $120,000 (2014 - 2018).
- PI - Trusted World of Corona (TWOC), PhD scholarship and joint supervision agreement with Radboud University Nijmegen, Health~Holland, €110,000 (2021 - 2023).

SELECTED PUBLICATIONS

112 peer-reviewed publications in international journals including Nature, Nature Communications, Frontiers in Genetics, Pharmacogenomics Journal and Human Molecular Genetics; 22 published since joining Northumbria (2022), 89% as corresponding author.

- Amegah AK, Kyobutungi C, **Chimusa ER**. Health data science revolution in Africa: potential for addressing the myriad health challenges. Nat Commun. 2025; 16:10919.
- Wonkam A, Esoh K, Levine RM, et al. FLT1 and other candidate fetal haemoglobin modifying loci in sickle cell disease in African ancestries. Nat Commun. 2025; 16:2092.
- Mugo JW, Mulder N, Chimusa ER. Data simulation to optimize frameworks for genome-wide association studies in diverse populations. Front Genet. 2025; 16:1559496.
- Agamah FE, Ederveen THA, Skelton M, Martin DP, **Chimusa ER,** 't Hoen PAC. Network-based multi-omics disease-drug associations reveal drug repurposing candidates for Covid-19 disease phases. DrugRepo. 2024;1(1).
- Thami PK, Choga WT, Dandara C, O'Brien SJ, Essex M, Gaseitsiwe S, **Chimusa ER**. Whole genome sequencing reveals population diversity and variation in HIV-1 specific host genes. Front Genet. 2023; 14:1290624.
- **Chimusa ER**, Defo J. Dissecting meta-analysis in GWAS era: Bayesian framework for gene/subnetwork-specific meta-analysis. Front Genet. 2022; 13:838518.

Books & Book Chapters

- Author - Mapping Genes Underlying Ethnic Differences in Admixed Populations. LAP LAMBERT Academic Publishing (2016), ISBN 978-3330010734, 236 pp.
- Chimusa ER. Computational disease-risk prediction: tools and statistical approaches. Academic Press (2025), pp. 91–106, ISBN 9780443185465.
- Plus, four further book chapters in bioinformatics and genomics methods (IntechOpen, Springer, Academic Press), including on local ancestry deconvolution, data-driven learning algorithms for post-genomic medicine, and pharmacogenomics data standards.

Link to full publication list https://orcid.org/0000-0001-8846-2047

SOFTWARE & COMPUTATIONAL TOOLS (SELECTED)

Leads development of open-source computational genomics tools (github.com/echimusa):

- FRANC - unified framework for multi-way local ancestry inference, integrating eight state-of-the-art ancestry deconvolution tools.
- FRACTALSIM - multi-scenario genome-wide population genetics simulation framework for modelling genetic architectures.
- PROXYANC - algorithms linking observed linkage disequilibrium in admixed populations to ancestral population differentiation.
- ancGWAS - graph-based method identifying disease-associated sub-networks underlying ethnic differences in admixed populations.
- ancMETA - Bayesian graph-based framework for gene/subnetwork-specific meta-analysis across multiple GWAS.
- A-DaGO-Fun - Gene Ontology semantic similarity-based functional analysis tool, widely used across the genomics and bioinformatics research community.

Lint to full list of developed Tools https://github.com/echimusa?tab=repositories

ACADEMIC LEADERSHIP & CITIZENSHIP

At Northumbria University (2022 - present)

2023 - present: Deputy Chair, Steering Group, Northeast Fatigue Research Network, UK.

2023 - present: Co-Chair, The State of Data Science for Health in Africa - a US-NIH, Wellcome, Bill & Melinda Gates Foundation initiative.

2024 - present: Deputy Chair, H3ABioNet - the pan-African Bioinformatics Network for Human Heredity and Health, spanning 28 institutions across 16 African countries.

2025 - 2026: Member, BBSRC Strategy Advisory Panel - shaping policy and strategy for UK bioscience research and innovation.

2025 - present: Mentor and supervision BCS Level 7 AI Data Specialist apprenticeship to Dr Ruchi Shukla (Assistant Professor); supporting grant development, bioinformatics capacity building.

2023 - present: Member, Race Equality Steering Group, Northumbria University; contributor to the University's first Race Equality Charter (Bronze) award (2024).

2022 - present: REF Assessor, Northumbria University.

2022 - present: Established and leading Northumbria's AWS-based bioinformatics Cloud infrastructure (in-silico functional computing / cloud lab).

2023 - 2024: Member of leading team of the development of the Northumbria University's 5-year High-Performance Computing strategy.

Previously, at the University of Cape Town

2016 - 2018: Deputy Chair, Health & Safety and Governance Committee, Institute of Infectious Disease and Molecular Medicine (IDM).

2017 - 2018: Bioinformatics Expert, H3Africa Consortium - co-designed the first African custom genotyping/sequencing array (the H3Africa Chip).

2019 - 2020: Chair, Equity, Diversity and Inclusion Steering Committee, IDM.

2021 - 2022: Deputy Chair, Advocacy for Transformation Committee, Faculty of Health Sciences.

2020 - 2022: Member, University of Cape Town Vision 2030 Taskforce - University strategy on embedding decoloniality, employment and promotion equity, anti-discrimination and inclusivity policy.

2014 - 2018: Consultant, Centre for Proteomics and Genomics Research (CPGR) Knowledge Transfer Programme - training and transferring 'omics' knowledge among researchers and faculty across the African continent (Chimusa et al. (2015) PLoS Comput Biol, 0.1371/journal.pcbi.1004512).

ESTEEM INDICATORS & HONOURS

2024 - Senior Fellow, Advance HE (2024).

2017 - C2 Rating, National Research Foundation (NRF), South Africa (2017) — peer-reviewed national rating recognising an established researcher with a sustained, internationally visible track record.

2022 - present: Elected Affiliate, The World Academy of Sciences (TWAS), Trieste, Italy (2017); Alumni Affiliate.

2020 - present: Elected Future Leader, Science & Technology in Society (STS) Forum, Japan (2018); Fellow.

2020 - present: Fellow, Next Einstein Forum.

2012 - 2014: Carnegie Corporation of New York PhD Scholarship and Postdoctoral Award, University of Cape Town (2012 – 2014)

2011 - Young Investigator Award, 12th International Congress of Human Genetics / The American Society of Human Genetics.

2010 - Prize, Intel Software Multi-core and Parallel Programming, Stellenbosch University.

PROFESSIONAL MEMBERSHIPS & EDITORIAL ROLES

- Editorial Board Member, Genes (**2026 - present**).
- Editorial Board Member, Frontiers in Oncology (**2019 - present**).

- Guest Editorial Board, Bioinformatics and Briefings in Bioinformatics (**2019 - 2021**).
- Associate Editorial Board Member, Bentham Science (**2017 - present**).
- Scientific Editorial Board, PLoS One (**2022 - present**).
- Co-Chair, State of Data Science Collection, Nature Portfolio journals (**2021 - present**).
- Member & Public Relations Officer, South African Society for Bioinformatics (SASBi) (**2014 - present**).
- Member, Southern African Society for Human Genetics (SASHG) (**2011 - present**).
- Member/Co-Investigator, Human Heredity and Health in Africa (H3Africa) (**2013 - present**).
- Member, International Society for Computational Biology (ISCB) (**2010 - present**).
- Member, European Conference on Computational Biology (ECCB) (**2013 - present**).
- Associate Member, Institute for Infectious Disease and Molecular Medicine (IDM), University of Cape Town (**2015 - present**).
- Member, North of England Genetic Epidemiology Group (NEGEG), UK (**2022 - present**).

GRANT REVIEW & ADVISORY PANELS

- Member, Selection Panel, Network Strengthening Grants scheme, Academy of Medical Sciences (**2025 - 2027**).
- Mentor, Academy of Medical Sciences Mentoring Programme, supporting a Researchers at Risk Fellow (**2025 - 2027**).
- Expert Panel Reviewer, European Innovation Council Accelerator (**2024 - 2025**).
- International Expert Reviewer & Panellist, Research Agency, Recovery and Resilience Plan of the Slovak Republic (**2024 - 2030**).
- International Expert Reviewer & Panellist, Latvian Council of Science; Slovak Research and Development Agency (**2023 - 2026**).
- International Expert Reviewer & Panellist, Research and Innovation Foundation, Cyprus (**2023 - 2027**).
- Panel Reviewer, BBSRC Doctoral Focal Awards (**2023 - present**).
- Expert/Panel Reviewer, European Commission HORIZON-JU (HORIZON-JU-GH-EDCTP) and ERASMUS-EDU (**2022 - present**).
- Grant Reviewer, Wellcome Trust (**2021 - present**).

INVITED KEYNOTE & PLENARY PRESENTATIONS (SELECTED, 2022 – 2026)

- "Future of bioinformatics and artificial intelligence: data integration, added value and current challenges" - NGS4-ECOPROD European Project Consortium, Tunis, Tunisia (**2026**).
- "How can bias in AI systems be addressed to ensure equitable outcomes?" - Commonwealth Scholars Connect & Collaborate Event, QEII Centre, Westminster, London (**2026**).
- "The State of Data Science for Health in Africa" - 5th DS-I Africa Consortium Meeting, Accra, Ghana (**2025**, Keynote & Chair).
- "Bias and AI-driven next generation disease risk prediction" - Institute of Medical Research and Medicinal Plants Studies, Yaoundé, Cameroon (**2025**).
- "Promising Dilemma Studies in Genetic Risk Prediction and Genome-Wide Association Studies" -18th TYAN International Thematic Workshop & 4th African Symposium on Big Data, Akure, Nigeria (**2025**).
- "Affordable AI-driven Wearable Biosensing network for disease early detection" - International Healthcare Facilities Management Symposium, Türkiye (**2024**, virtual).
- "Tantalizing Dilemma in Genetics Risk Prediction: Case of African Genomics" - Curating the Clinical Genome Conference, Wellcome Genome Campus, London (**2022**).

TEACHING & PROGRAMME LEADERSHIP

- Programme Lead, MSc Bioinformatics and AI-driven Omics, Northumbria University (**launching September 2027**).
- Over 12 years' HE teaching experience across undergraduate and postgraduate bioinformatics and computational biology programmes
- Led curriculum development and South African Council on Higher Education accreditation of the Honours/MSc/PhD Computational Health Informatics programme, University of Cape Town.
- Invited teaching and training: Harvard T.H. Chan School of Public Health; African Institute for Mathematical Sciences (Cape Town, Kigali, Dar es Salaam, Cameroon, Ghana, Senegal); SFI Centre for Research Training in Genomics Data Science, University of Galway, Ireland.
- Learning & Teaching Mentor for colleagues undertaking PGCAP / NUPRS Fellowship routes.
- Led design of diverse, inclusive assessment strategies (formative, summative, oral and practical) across taught BSc/MSc modules; introduced real-time feedback tools in direct response to student input.
- Identified modules with higher student disengagement risk and led targeted intervention; mentored module tutors on fair, consistent and transparent assessment practice, including appeals and re-assessment procedures.

RESEARCH STUDENT SUPERVISION

11 PhD students supervised to completion (lead or co-supervisor); 1 current PhD student at Northumbria University. Graduates have progressed to postdoctoral and faculty positions at institutions including the NIH (Bethesda, USA), Johns Hopkins University School of Medicine, and universities across Africa, Europe and North America.

- Current: Nefeli Tzanaki, Northumbria University (ongoing)
- Completed (selected): Francis Agamah (2024, Postdoctoral Fellow, University of Cape Town); Jacquiline Wangui Mugo (2023, Postdoctoral Fellow, UCT Lung Institute); Abdoulaye Yalcouye (2022, Postdoctoral Fellow, Johns Hopkins University School of Medicine); Edmond Tingang Wonkam (2021, Postdoctoral Fellow, NIH, Bethesda); Delesa Damena Mulisa (2020, Postdoctoral Fellow, NIH, Bethesda); Denis Awany (2020, Postdoctoral Fellow, South African Tuberculosis Vaccine Initiative).
- Also supervised over a dozen MSc/Honours students, several progressing to PhD study or bioinformatics careers internationally, including at the NIH and King Saud bin Abdulaziz University for Health Sciences.
- Selected PhD external examiner: University of the Witwatersrand; Makerere University; University of Western Cape.

PROFESSIONAL DEVELOPMENT & CERTIFICATION

- Certificate, New Frontiers Phase 1 Entrepreneur Development Programme, Synergy Centre, Technological University Dublin, Tallaght Campus, Ireland (**2021**).
- Professional Diploma, AWS Solutions Architect (SAA-C02) - Designing Secure Applications & Architectures (2022).
- Certificate, Business Development Programme, Europe's First University Venture Competition, Hietzing, Vienna (2023).

KNOWLEDGE EXCHANGE & COMMERCIAL IMPACT

- Data Licence Agreement (in progress) for commercialisation of a large-scale liver cancer genomic dataset - Northumbria University, Cancer Research Horizons and Newcastle University.

- Established AWS-based bioinformatics Cloud infrastructure at Northumbria (in-silico functional computing / cloud lab).
- Active contributor to NIH-funded pan-African genomics networks: H3ABioNet, H3Africa, and DS-I Africa.